

|                                        |           |             |
|----------------------------------------|-----------|-------------|
| Name: Flavia María Graniello Álvarez   |           |             |
| School: Academica Británica Cuscatleca |           |             |
| Group: 4th quin...                     |           | Sex: Female |
| Date of first test: 13/11/2024         | Level: 8B | Age: 8:06   |
| Date of second test: 10/10/2025        | Level: 9B | Age: 9:05   |

Scores

| No. attempted (/54) | SAS | SAS (with 90% confidence bands) |    |    |    |     |                                   |     |     |     |  | Overall ST | PR | GR (/25) | End of KS2 indicator | Progress Category |
|---------------------|-----|---------------------------------|----|----|----|-----|-----------------------------------|-----|-----|-----|--|------------|----|----------|----------------------|-------------------|
|                     |     | 60                              | 70 | 80 | 90 | 100 | 110                               | 120 | 130 | 140 |  |            |    |          |                      |                   |
| 53                  | 108 |                                 |    |    |    |     | <div><div></div><div></div></div> |     |     |     |  | 6          | 70 | =3       | 107                  | Much higher       |

Previous & Current Scores

| Date       | Age at test (yrs:mths) | Level / Form | No. attempted | SAS | SAS (with 90% confidence bands) |    |    |    |     |     |     |     |     |  | ST | PR | End of KS2 indicator |
|------------|------------------------|--------------|---------------|-----|---------------------------------|----|----|----|-----|-----|-----|-----|-----|--|----|----|----------------------|
|            |                        |              |               |     | 60                              | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |  |    |    |                      |
| 07/05/2025 | 9:00                   | 9A           | 53 / 54       | 103 |                                 |    |    |    |     |     |     |     |     |  | 5  | 58 | 105                  |
| 10/10/2025 | 9:05                   | 9B           | 53 / 54       | 108 |                                 |    |    |    |     |     |     |     |     |  | 6  | 70 | 107                  |

Analysis of Curriculum content categories

| Curriculum content category | Number of questions | Student % correct | Standardisation % correct | Student / standardisation difference |
|-----------------------------|---------------------|-------------------|---------------------------|--------------------------------------|
| Number                      | 38                  | 78%               | 62%                       | 16%                                  |
| Measurement                 | 6                   | 50%               | 47%                       | 3%                                   |
| Geometry                    | 4                   | 50%               | 40%                       | 10%                                  |
| Statistics                  | 6                   | 67%               | 63%                       | 4%                                   |

Analysis of Process categories

| Process category                    | Number of questions | Student % correct | Standardisation % correct | Student / standardisation difference |
|-------------------------------------|---------------------|-------------------|---------------------------|--------------------------------------|
| Fluency in facts and procedures     | 13                  | 77%               | 64%                       | 13%                                  |
| Fluency in conceptual understanding | 22                  | 75%               | 64%                       | 11%                                  |
| Problem solving                     | 7                   | 33%               | 34%                       | -1%                                  |
| Mathematical reasoning              | 12                  | 80%               | 57%                       | 23%                                  |

## Implications for teaching and learning

- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
  - Flavia María took *PTM8B* in November 2024 and from then until now has made much higher than expected progress in maths.
- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- Where performance is fairly evenly balanced across curriculum categories, this suggests that Flavia María will generally demonstrate a level of understanding of mathematical concepts appropriate with this age, irrespective of whether this is in a format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Fluency and agility are equally well developed in both written and mental formats and Flavia María is developing the language of mathematics in line with expectations for her age group.
- Where performance across the curriculum categories are uneven, specific areas of weakness might be addressed as follows:
  - Further targeted practice in the areas identified as being relatively weaker.
  - Practical activities using equipment that is designed to help Flavia María to 'see' the thinking that lies behind any concepts that are not yet secure.
  - Get Flavia María to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Flavia María is generally secure in performing the basic mental calculations expected for this age group. These include age-appropriate fluency with whole numbers, fractions, decimals, percentages and ratio as well as an understanding of the number system, place value and the connections between inverse operations such as multiplication and division.
- Flavia María is developing mathematical reasoning and the ability to solve a wider range of problems, including increasingly complex properties of numbers and arithmetic, and problems demanding efficient written and mental methods of calculation. Flavia María has been introduced to the language of algebra to extend problem solving to a wider variety of contexts and is also developing an understanding of the geometry of increasingly complex shapes and the language needed to describe these.
- Next steps should include opportunities to stretch and develop conceptual understanding, to increase mental agility and to progress the development of formal written mathematics using age-appropriate symbols and methodologies. This can be done by emphasising the higher order skills of hypothesising or predicting (What is an approximate value for  $49 \times 11$ ?); designing and comparing procedures (How many different ways can we work out  $53 - 18$ ? Which is the most efficient?); interpreting results (If  $9 \times 5 = 45$ , how many other calculations can we easily work out, e.g.  $0.9 \times 5$ ?) and applying reasoning (If I have 39p in my purse, what coins might they be?). In the formal written development there should be an emphasis on developing the correct use of mathematical language, for example in using ':' for ratio and the '=' sign correctly as a mathematical sentence.
- By providing practice time and frequent opportunities for Flavia María to use one or more known facts to work out more facts, and by engaging Flavia María in discussion and providing opportunities to explain methods and strategies to you and her peers, Flavia María will be helped to make the connections between mental calculations and the written explanations. This will help to develop both problem solving abilities and the language of mathematics, thus improving both mental agility and the ability to plan and carry out accurate calculations in a more formal way.

|                                               |                  |                    |
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## Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

| Question number | Curriculum category | Process category                    | Question content                                                                                           | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| MM1             | Number              | Fluency in facts and procedures     | What is six doubled?                                                                                       | 1 / 1      | 68              | 87                        |
| MM2             | Number              | Fluency in facts and procedures     | Multiply three by seven.                                                                                   | 1 / 1      | 52              | 63                        |
| MM3             | Number              | Fluency in conceptual understanding | Write in figures the number four thousand and thirty-six.                                                  | 1 / 1      | 36              | 63                        |
| MM4             | Number              | Mathematical reasoning              | A packet of forty-five sweets is shared equally between five people. How many sweets does each person get? | 0 / 1      | 24              | 64                        |
| MM5             | Number              | Fluency in facts and procedures     | A cake is cut into thirds. How many pieces are there?                                                      | 1 / 1      | 64              | 56                        |
| MM6             | Measures            | Fluency in conceptual understanding | My watch says half past two in the afternoon. What would a twenty four hour digital watch say?             | 0 / 1      | 4               | 28                        |
| MM7             | Number              | Mathematical reasoning              | One car has four tyres. How many tyres do three cars have?                                                 | 1 / 1      | 72              | 80                        |
| MM8             | Number              | Mathematical reasoning              | There are six eggs in one box. How many eggs are there in seven boxes?                                     | 0 / 1      | 8               | 52                        |
| MM9             | Number              | Fluency in conceptual understanding | What is twenty-five take away thirteen?                                                                    | 1 / 1      | 64              | 66                        |
| MM10            | Number              | Fluency in facts and procedures     | Write a multiple of six that is between fifteen and twenty-five.                                           | 0 / 1      | 16              | 44                        |
| MM11            | Number              | Fluency in conceptual understanding | Look at these numbers. Write down all the odd numbers.                                                     | 1 / 1      | 52              | 64                        |
| MM12            | Number              | Fluency in facts and procedures     | Add these numbers: forty-one, twenty-five and fourteen.                                                    | 1 / 1      | 32              | 43                        |
| MM13            | Number              | Mathematical reasoning              | There are thirty-two people in a restaurant. Thirteen are adults. How many are children?                   | 1 / 1      | 28              | 36                        |
| MM14            | Measures            | Mathematical reasoning              | I have twenty-seven pence. I want to spend thirty-five pence. How much more money do I need?               | 1 / 1      | 60              | 61                        |
| MM15            | Measures            | Fluency in facts and procedures     | How many days are there in eight weeks?                                                                    | 0 / 1      | 16              | 47                        |
| AU1             | Number              | Fluency in conceptual understanding | Draw lines to connect the matching fractions.                                                              | 2 / 2      | 86              | 89                        |
| AU2             | Number              | Fluency in conceptual understanding | What fraction of this shape is shaded?                                                                     | 1 / 1      | 36              | 78                        |
| AU3a            | Number              | Fluency in conceptual understanding | Which day had the lowest midnight temperature?                                                             | 1 / 1      | 56              | 59                        |

| Question number | Curriculum category | Process category                    | Question content                                                                                                             | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| AU3b            | Number              | Fluency in conceptual understanding | Which day had a midnight temperature 1 degree higher than Friday?                                                            | 1 / 1      | 40              | 50                        |
| AU4             | Number              | Mathematical reasoning              | On Sunday the midnight temperature was -2 degrees. The next day, the midnight temperature was 5 degrees higher. What was it? | 1 / 1      | 48              | 49                        |
| AU5             | Geometry            | Fluency in facts and procedures     | Name the 3 shapes that are labelled.                                                                                         | 1 / 1      | 28              | 46                        |
| AU6a            | Geometry            | Fluency in facts and procedures     | In which labelled shape are all the angles right angles?                                                                     | 1 / 1      | 48              | 60                        |
| AU6b            | Geometry            | Fluency in facts and procedures     | In which labelled shape are all the angles greater than a right angle?                                                       | 0 / 1      | 32              | 42                        |
| AU7             | Number              | Mathematical reasoning              | Here are 3 more fraction sums that make 1 whole. Fill in the missing numbers.                                                | 2 / 2      | 22              | 62                        |
| AU8             | Measures            | Mathematical reasoning              | Write the times in the correct boxes.                                                                                        | 2 / 2      | 44              | 36                        |
| AU9             | Statistics          | Fluency in conceptual understanding | Put a circle round the country that has the least tigers.                                                                    | 1 / 1      | 64              | 83                        |
| AU10            | Statistics          | Fluency in conceptual understanding | Roughly how many tigers are there in Malaysia?                                                                               | 1 / 1      | 76              | 81                        |
| AU11            | Statistics          | Fluency in conceptual understanding | Put a circle round the 3 countries that have less than 500 tigers.                                                           | 1 / 1      | 68              | 70                        |
| AU12a           | Number              | Fluency in conceptual understanding | Which number has 7 as its hundreds digit?                                                                                    | 0 / 1      | 60              | 83                        |
| AU12b           | Number              | Fluency in conceptual understanding | Which number is 10 times bigger than 62?                                                                                     | 1 / 1      | 40              | 63                        |
| AU12c           | Number              | Fluency in conceptual understanding | Which number is not a whole number?                                                                                          | 0 / 1      | 60              | 70                        |
| AU12d           | Number              | Fluency in conceptual understanding | Which is the smallest number?                                                                                                | 1 / 1      | 92              | 81                        |
| AU13            | Number              | Fluency in facts and procedures     | Here is a number machine. Fill in the missing number.                                                                        | 1 / 1      | 84              | 74                        |
| AU14a           | Number              | Mathematical reasoning              | Fill in the missing number.                                                                                                  | 1 / 1      | 68              | 66                        |
| AU14b           | Number              | Problem solving                     | What number comes out the same as it goes in?                                                                                | 1 / 1      | 28              | 40                        |
| AU15a           | Number              | Fluency in facts and procedures     | Which is the biggest number?                                                                                                 | 1 / 1      | 88              | 94                        |
| AU15b           | Number              | Fluency in facts and procedures     | Write in a number that is bigger than the biggest number here.                                                               | 1 / 1      | 92              | 86                        |
| AU16a           | Number              | Fluency in facts and procedures     | Which are the smallest two numbers in the list?                                                                              | 1 / 1      | 100             | 93                        |
| AU16b           | Number              | Mathematical reasoning              | Write in any whole number that is between the 2 smallest numbers.                                                            | 1 / 1      | 64              | 74                        |
| AU17            | Measures            | Fluency in conceptual understanding | Which two of these are equal to seventy pence?                                                                               | 1 / 2      | 54              | 68                        |
| AU18            | Number              | Fluency in conceptual understanding | What number is exactly between 41 and 55?                                                                                    | 1 / 1      | 36              | 31                        |
| AU19            | Number              | Fluency in conceptual understanding | What number is exactly between 4.5 and 5.5?                                                                                  | 1 / 1      | 48              | 51                        |
| AU20            | Number              | Fluency in conceptual understanding | What number is exactly halfway between $\frac{2}{7}$ and $\frac{4}{7}$ ?                                                     | 1 / 1      | 56              | 71                        |

| Question number | Curriculum category | Process category                    | Question content                                                                          | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|-------------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| AU21            | Geometry            | Problem solving                     | Write the letters of each shape into the correct boxes.                                   | 1 / 3      | 28              | 31                        |
| AU22            | Number              | Mathematical reasoning              | Now choose 4 different numbers to show more multiplications with the answer 32.           | 1 / 2      | 36              | 47                        |
| AU23            | Measures            | Fluency in conceptual understanding | At the bank, Mr Smith changes a £10 note for 20p pieces. How many 20p pieces does he get? | 0 / 1      | 16              | 29                        |
| AU24            | Number              | Mathematical reasoning              | What is the smallest possible number you can make using all 4 of these cards?             | 1 / 1      | 76              | 84                        |
| AU25            | Number              | Problem solving                     | Arrange the cards to make 2 numbers whose sum is 74.                                      | 1 / 1      | 48              | 57                        |
| AU26            | Number              | Problem solving                     | Arrange the cards to make 2 numbers whose difference is 26.                               | 0 / 1      | 32              | 35                        |
| AU27a           | Number              | Problem solving                     | How many stickers will be left when she has made all the cards?                           | 0 / 1      | 16              | 21                        |
| AU27b           | Number              | Problem solving                     | How many more invitations can she make with the left-over stickers?                       | 0 / 1      | 8               | 24                        |
| AU28            | Statistics          | Fluency in conceptual understanding | How many babies were at the medical centre on that Wednesday?                             | 0 / 1      | 20              | 40                        |
| AU29            | Statistics          | Fluency in conceptual understanding | Put an x on the graph in question 17 to show John's heartbeat.                            | 1 / 1      | 56              | 70                        |
| AU30            | Statistics          | Problem solving                     | Which 2 of these age groups could Erin be in?                                             | 0 / 1      | 8               | 33                        |